INFINITE LIMITS, CONTINUITY

Def'n

$\lim_{x \to \infty} f(x) = L$ if for all $\varepsilon > 0$ there exists $N \in \mathbb{R}$ such that if $x > N$, $|f(x) - L| < \varepsilon$

$\lim_{x \to -\infty} f(x) = L$ if for all $\varepsilon > 0$ there exists $N \in \mathbb{R}$ such that if $x < N$, $|f(x) - L| < \varepsilon$.

Def'n

If $\lim_{x \to \infty} f(x) = L$ or $\lim_{x \to -\infty} f(x) = L$, then L is a <u>horizontal asymptote</u> of f.

Def'n

$\lim_{x \to \infty} f(x) = \infty$ if for all $M > 0$, there exists $N \in \mathbb{R}$ such that if $x > N$, $f(x) > M$.

$\lim_{x \to \infty} f(x) = -\infty$ if for all $M > 0$, there exists $N \in \mathbb{R}$ such that if $x > N$, $f(x) < M$.

$\lim_{x \to -\infty} f(x) = \infty$ if for all $M > 0$, there exists $N \in \mathbb{R}$ such that if $x < N$, $f(x) > M$.

$\lim_{x \to -\infty} f(x) = -\infty$ if for all $M > 0$, there exists $N \in \mathbb{R}$ such that if $x < N$, $f(x) < M$.

Theorem

If $g(x) \leq f(x) \leq h(x)$ for all $x \geq N$ for some $N \in \mathbb{R}$ and $\lim_{x \to \infty} g(x) = L = \lim_{x \to \infty} h(x)$, then $\lim_{x \to \infty} f(x) = L$.

Proposition

In general, for $f(x) = \dfrac{a_n x^n + \cdots + a_1 x + a_0}{b_m x^m + \cdots + b_1 x + b_0}$,

$$\lim_{x \to \pm\infty} f(x) = \begin{cases} \dfrac{a_n}{b_m} & \text{if } n = m \\ 0 & \text{if } n < m \\ \text{DNE} & \text{if } n > m \end{cases}$$

Theorem

Fundamental Logarithm Limit

$$\lim_{x \to \infty} \frac{\ln(x)}{x^p} = 0 \quad \text{and} \quad \lim_{x \to \infty} \frac{x^p}{e^x} = 0$$

for any $p > 0$.

Corollary $(\ln(x))^p \ll x^p \ll p^x \ll x^x$ as $x \to \infty$ for $p > 0$

Def'n $\lim_{x \to a^+} f(x) = \infty$ if for all $M > 0$ there exists $\delta > 0$ such that if $a < x < a + \delta$, $f(x) > M$.

$\lim_{x \to a^-} f(x) = \infty$ if for all $M > 0$ there exists $\delta > 0$ such that if $a - \delta < x < a$, $f(x) > M$.

$\lim_{x \to a^+} f(x) = -\infty$ if for all $M > 0$, there exists $\delta > 0$ such that if $a < x < a + \delta$, $f(x) < M$.

$\lim_{x \to a^-} f(x) = -\infty$ if for all $M > 0$, there exists $\delta > 0$ such that if $a - \delta < x < a$, $f(x) < M$.

Def'n If $\lim_{x \to a^+} f(x) = \infty = \lim_{x \to a^-} f(x)$, then $x = a$ is a <u>vertical asymptote</u> of f.

Def'n #1 A function f is continuous at $x = a$ if:

1) $\lim_{x \to a} f(x)$ exists
2) $\lim_{x \to a} f(x) = f(a)$

Def'n #2 A function f is continuous at $x = a$ if for all $\varepsilon > 0$, there exists $\delta > 0$ such that if $|x - a| < \delta$, $|f(x) - f(a)| < \varepsilon$.

Def'n #3 A function f is continuous at $x = a$ if $\{x_n\}$ is a sequence where if $\lim_{n \to \infty} x_n = a$, $\lim_{n \to \infty} f(x_n) = f(a)$

Theorem If $f(x)$ is invertible, $f(a) = b$, and f is continuous at $x = a$, then $f^{-1}(x)$ is continuous at $x = b$.

Theorem Let f and g be continuous at $x = a$. Then,

1) $f + g$ is continuous at $x = a$
2) fg is continuous at $x = a$.
3) If $g \neq 0$, then $\frac{f}{g}$ is continuous at $x = a$.

Theorem If f is continuous at $x = a$ and g is continuous at $x = f(a)$, then $h = g \circ f$ is continuous at $x = a$.

Def'n f is continuous on (a, b) (or $\mathbb{R}$) if it is continuous for all $x \in (a, b)$ (or $\mathbb{R}$).

Def'n f is continuous on $[a, b]$ if:

1) f is continuous on (a, b)
2) $\lim_{x \to a^+} f(x) = f(a)$
3) $\lim_{x \to b^-} f(x) = f(b)$

QUIZ #5, PRACTICE PROBLEMS #5

PRACTICE PROBLEMS

1. (a)

$$= \lim_{x \to \infty} \frac{x\left(\frac{\sqrt{x}}{x} + \frac{\ln(x)}{x} - \frac{x}{x}\right)}{1 - 2x}$$

$$= \lim_{x \to \infty} \frac{x\left(\frac{\sqrt{x}}{x} + \frac{\ln(x)}{x} - 1\right)}{x\left(\frac{1}{x} - \frac{2x}{x}\right)}$$

$$= \lim_{x \to \infty} \frac{\frac{\sqrt{x}}{x} + \frac{\ln(x)}{x} - 1}{\frac{1}{x} - 2}$$

$$= \frac{0 + 0 - 1}{0 - 2}$$

$$= \frac{1}{2}$$

(b)

$$= \lim_{x \to \infty} e^{-x} - \frac{x(e^{x})^{\frac{1}{2}}}{e^{x}}$$

$$= \lim_{x \to \infty} e^{-x} - \left(\frac{x^2}{e^{x}}\right)^{\frac{1}{2}}$$

$$= 0 - 0^2$$

$$= 0$$

(c)

$$= \lim_{x \to \infty} \frac{\ln(x)}{x^p}\left(\frac{e^{x}}{x^p}\right)$$

$$= \lim_{x \to \infty} \frac{\ln(x)}{x^{2p}}\left(e^{x}\right)$$

$$= \infty$$

(d)

$$= \lim_{x \to \infty} \left(\frac{\ln(x)}{x^{\frac{1}{e}}}\right)^{e}$$

$$= 0^{e}$$

$$= 0$$

2. <u>Horizontal Asymptotes</u>: Let $x \to \pm\infty$. Since $-2x^2 + 2 \to -\infty$ $x \to \pm\infty$, we have

$$\lim_{x \to \infty} \frac{1}{-2x^2 + 2} = 0 \quad \text{and} \quad \lim_{x \to -\infty} \frac{1}{-2x^2 + 2} = 0$$

Thus, $y = 0$ is a horizontal asymptote of $f(x)$.

<u>Vertical Asymptotes</u>: Since $f(x)$ is defined everywhere except when $-2x^2 + 2 = 0$, i.e. $x = \pm 1$

Note that if $\lim_{x \to 1^+} f(x) = -\infty$ and $\lim_{x \to 1^-} f(x) = \infty$

Similarly, $\lim_{x \to -1^+} f(x) = \infty$ and $\lim_{x \to -1^-} f(x) = -\infty$.

Thus, $x = \pm 1$ are vertical asymptotes of $f(x)$.

3. Note that $f(2) = 2(2)^2 + 9 = 17$. Then, for all $\varepsilon > 0$ there exists $\delta > 0$ such that if $|x - 2| < \delta$, $|(2x^2 + 9) - 17| < \varepsilon$.

$$|(2x^2 + 9) - 17| = |2x^2 - 8| = |2(x-2)(x+2)| < \varepsilon$$

Choose $\delta < 1$. Then,

$$|x - 2| < 1 \Rightarrow 2 - 1 < x < 2 + 1 \Rightarrow 1 < x < 3$$

Since $1 < x < 3$, we have

$$6 < 2|x + 2| < 10 \Rightarrow 2|x+2||x-2| < 10|x-2|$$

Let $\delta = \min\left\{1, \frac{\varepsilon}{10}\right\}$. Then,

$$|2(x+2)(x-2)| < 10|x-2| < 10\delta \leq \frac{10\varepsilon}{10} = \varepsilon$$

It follows that $\lim\limits_{x \to 2} (2x^2 + 9) = 17$.

4. Note that $\dfrac{x^2 - 4}{x^2 + x - 6}$ is continuous everywhere except when $x^2 + x - 6 = 0$, i.e. $x = -3, 2$.

If $x \to -3$, $\lim\limits_{x \to -3} \dfrac{x^2 - 4}{x^2 + x - 6} \cos(x^2) = \dfrac{5}{0} \cos(9)$

Thus, $\lim\limits_{x \to -3} \dfrac{x^2 - 4}{x^2 + x - 6} \cos(x^2)$ DNE.

If $x \to 2$, $\lim\limits_{x \to 2} \dfrac{x^2 - 4}{x^2 + x - 6} \cos(x^2) = \dfrac{0}{0} \cos(4)$

Thus, $\lim\limits_{x \to 2} \dfrac{x^2 - 4}{x^2 + x - 6} \cos(x^2)$ exists but $\neq f(a)$

Since f is not continuous at $x = -3, 2$, f is continuous at $x \in (-\infty, -3) \cup (-3, 2) \cup (2, \infty)$

5. f is continuous everywhere except possibly at $x = 2$.
For f to be continuous at $x = 2$,

$$\lim_{x \to 2} f(x) = f(2) = 8 - 2c$$

Observe that $\lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} cx^2 + 2x = 4c + 4$

For $\lim_{x \to 2} f(x)$ to exist, we need

$$8 - 2c = 4c + 4 \Rightarrow -6c = -4 \Rightarrow c = \frac{2}{3}$$

It follows that $f(x)$ is continuous when $c = \frac{2}{3}$

6. (a) We are trying to prove $\lim_{x \to a} f(x) = f(a) = \lim_{h \to 0} f(a+h)$.

$$\begin{aligned}
\lim_{h \to 0} f(a+h) &= \lim_{h \to 0} f(a) + f(h) \\
&= \lim_{h \to 0} f(a) + \lim_{h \to 0} f(h) \\
&= f(a) + \lim_{h \to 0} f(h) \\
&= f(a) + f(0) \\
&= f(a+0) \\
&= f(a)
\end{aligned}$$

(b) We are trying to prove $\lim_{x \to a} f(x) = f(a) = \lim_{h \to 0} f(a+h)$.

$$\begin{aligned}
\lim_{h \to 0} f(a+h) &= \lim_{h \to 0} f(a) \cdot f(h) \\
&= \lim_{h \to 0} f(a) \cdot \lim_{h \to 0} f(h) \\
&= f(a) \cdot \lim_{h \to 0} f(h) \\
&= f(a) \cdot f(0) \\
&= f(a+0) \\
&= f(a)
\end{aligned}$$

7 (a), (e), (g)

8. (a) T
(b) T
(c) T
(d) T
(e) F